



Figure 1: Taxonomy of malware

Persistence mechanism: Malware is highly persistent and there are 'n' number of ways to infect Windows based machines. Identifying highly persistent malware is a lot tougher than the traditional malware.

Malware with multiple functions

Malware can possess two or more functionalities with equal threat and persistent levels, and such malware is identified as 'hybrid'.

For example,

- Trojan + Ransom = Hybrid Trojan
- Trojan + RAT = RAT
- P2P + Worm = Worm

Steps to building a malware analysis toolkit

The three steps needed to build an analysis toolkit are described below.

Create and allocate a safe environment for analysis:

The conventional way of examining malicious programs involves infecting a system with the malware and learning about its behaviour using appropriate monitoring tools. This requires a sandbox or virtual machine (VM) in order to investigate the behaviour without affecting the real production bed.

Sandbox the system from the production environment:

Running multiple VMs inside the sandbox helps to analyse the malware which seeks replication or interaction with other systems. Another useful feature of a VM is that an instant snapshot of the system can be recorded along with the system states (before and after infection of the system). In sandboxing, you can install as much RAM and memory, because some intelligent malware can detect its running state inside the VM.

Install a behaviour analysis tool: Before testing the malware in the sandbox, it is preferable to install



Figure 2: Splash screen of REMnux

appropriate monitoring tools, some of which are discussed in this article.

REMnux: A Linux reverse engineering toolkit

This article is primarily about the REMnux distro. This is a free Linux toolkit used for reverse engineering malicious software. It is bundled with various forensic investigation tools. The distro is based on Ubuntu, and it analyses both Linux and Windows based malware, examining obfuscated code, suspicious documents, etc. It can be freely downloaded from <https://remnux.org/> as an OVA file, which can be imported directly into any virtual machine. A Docker image is also available for this distro in the same link.

Let's now go through a tutorial on inspecting executables and file properties using REMnux.

Pescanner: To inspect any files, malicious software or suspicious content in the form of portable executables like *.exe*, *.msi*, *.so*, etc, use the following command:

```
pescanner <filename>
```

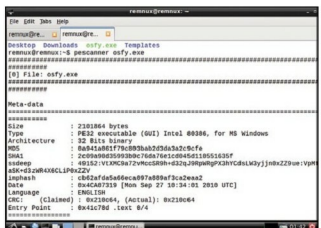


Figure 3: Screenshot of Pescanner - metadata information